Quiz instructions

Get a pen out now, before we hand out the quiz.

Leave the quiz facing down until we say it is time to start.

There are two versions of the quiz.

The people either side of you must have a different coloured quiz to you.

When we say it is time to stop, hand your quiz to the end of your row.

Use blue or black ink.

Make sure you write your name and SID.

This is a closed-book, closed-note quiz.

No electronic devices are allowed on your desk during the quiz.

Note your responses by completely filling in the relevant circle: ●

If you make a mistake, put an X through the circle: ⊗

COMP 4446 / 5046 Lecture 4: Inference – Greedy and Search

Jonathan K. Kummerfeld

Semester 1, 2025

THE UNIVERSITY OF SYDNEY

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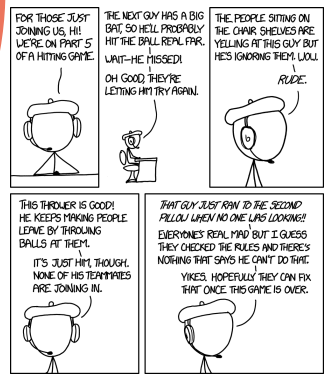
Play-By-Play



[The thrower started hitting the bats too much, so the king of the game told him to leave and brought out another thrower from thrower jail.]

Source: https://xkcd.com/1593/

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Exhaustive

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Now, let's explore different inference methods

Data	Examples of the language phenomena we want our system to handle
Model	A function that maps (input, output) pairs to scores
Inference Method	A way to make a prediction for an example given a Model
Metric	A function that gives a score to the output produced by a Model given some Data
Learning Method	A way to update a Model given Data a Metric and an Inference Method

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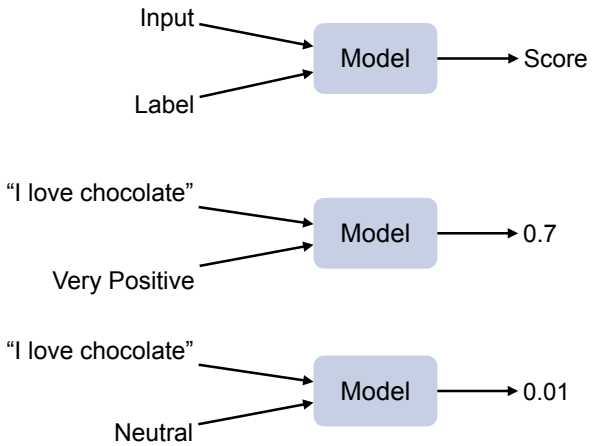
We'll treat the model as a black box



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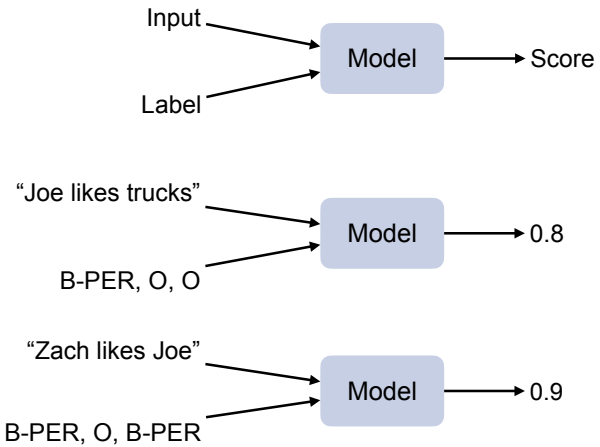
The model could score whole sequences



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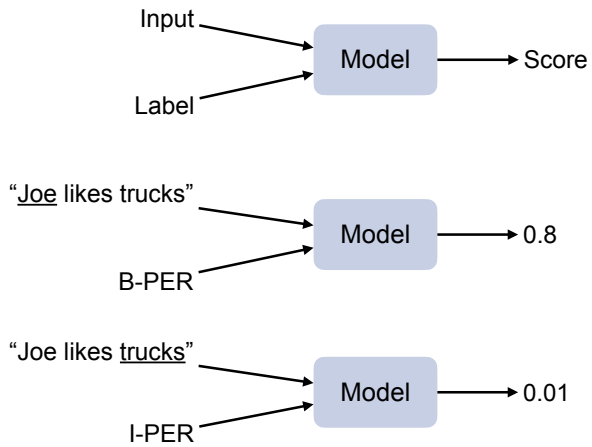
The model could score whole sequences, or just one part



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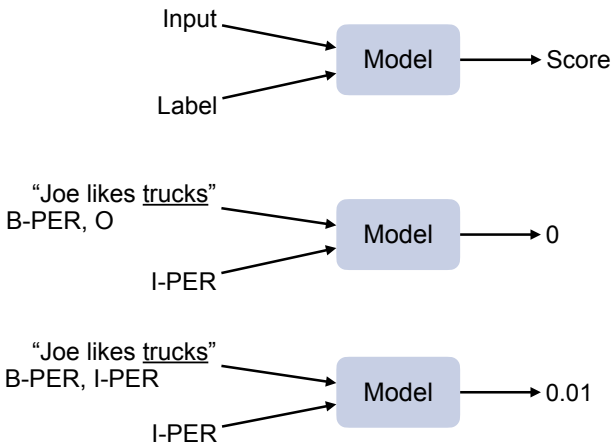
The model could score whole sequences, or just one part, possibly with context



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We'll use several running examples in this section

Noun	Verb	Prep	Noun
Adj	Noun	Verb	Noun
Fruit	flies	like	bananas

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We'll use several running examples in this section

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering
4. Take off heat and let cool completely (~20 min)
5. Return to stove and heat to desired temperature
6. Top with chocolate shavings

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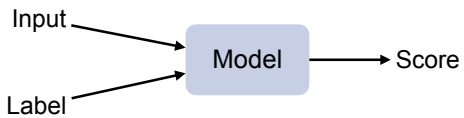
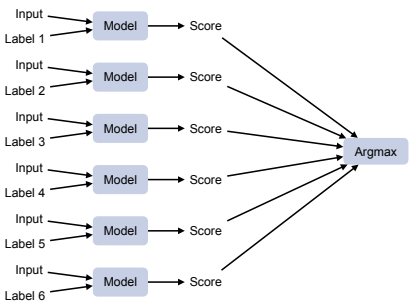
In assignment 2, we are using an exhaustive method





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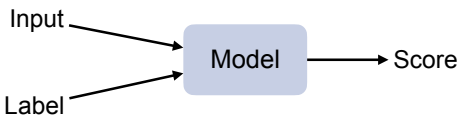

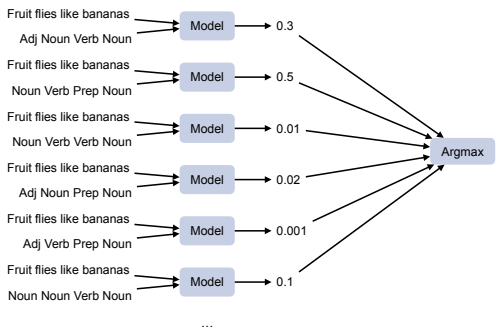

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Exhaustive search is flexible, but not scalable

Benefit - The model can look at the entire structure

Fruit flies like bananas
Adj Noun Verb Noun
→ Model → 0.3

Problem - for many tasks, the search space is huge

17 tags in Universal Dependencies

Options = $|\text{tags}|^{|\text{words}|}$

For this example, 17^4 = 83,521 options!

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Note: we are looking for the highest scoring output, which might be wrong

Fruit flies like bananas
Adj Noun Verb Noun
→ Model → 0.3

Fruit flies like bananas
Noun Verb Prep Noun
→ Model → 0.5

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This suggests another way of describing these two components of an NLP system

Model — Tries to assign the highest score to the true output

Inference Method — Tries to find the output that gets the highest score from the Model

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Greedy

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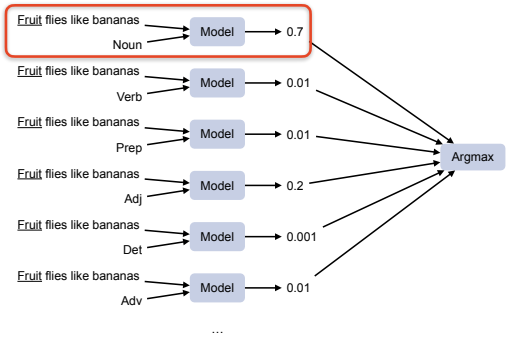
Core greedy idea: Make choices one at a time



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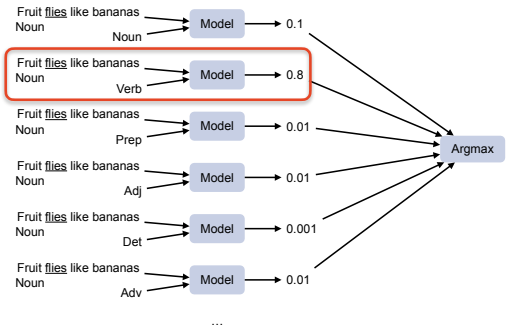
Core greedy idea: Make choices one at a time



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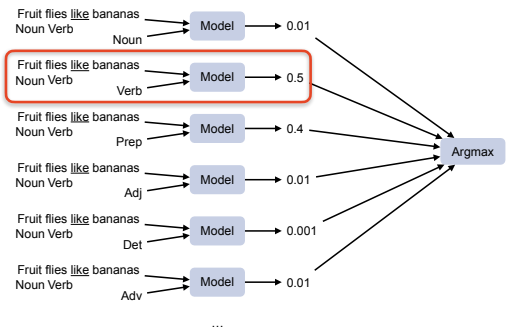
Core greedy idea: Make choices one at a time



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The complexity problem is fixed!

17 tags in Universal Dependencies

Options considered in each step = |tags|

Step = |words|

Complexity = |tags| * |words|
= 17 * 4
= 68

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But… the answer was different

Exhaustive:	Noun	Verb	Prep	Noun
Greedy:	Noun	Verb	Verb	Noun
	Fruit	flies	like	bananas

Note - both of these are wrong, but one is the highest scoring according to the model and the other is not.

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A few notes on comparing exhaustive and greedy

Sometimes the answer can match. For example, if every part of the output is independent.

Greedy has less information about the output, but can still use a lot of context to make the decision

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First variant: **Top-1**

Method: At each step, choose the highest scoring option

This is what we just saw!

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First variant: **Top-1**

Method: At each step, choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering
4. Melt chocolate on stove
5. Add milk
6. Heat
7. Melt chocolate on stove

…

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Second variant: **Random sampling**

Method: At each step, choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Melt chocolate on stove

Take off heat and let cool completely (~20 min)

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

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Third variant: **Top-K** sampling

Method: At each step, filter to the top K options, then choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

K = 3

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Third variant: **Top-K** sampling

Method: At each step, filter to the top K options, then choose using random sampling from the probability distribution

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

Pour into mug

Take off heat and let cool completely (~20 min)

0.38 - Melt chocolate on stove
0.37 - Take off heat and let cool completely (~20 min)
0.25 - Pour into mug
0.1 - Simmer for 5 minutes
…

K = 3

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Fourth variant: **Top-P** sampling

Method: At each step, filter to the options that cover P% of the probability distribution, then choose using random sampling

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

P = 80%

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Fourth variant: **Top-P** sampling

Method: At each step, filter to the options that cover P% of the probability distribution, then choose using random sampling

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Pour into mug

Steps:

Simmer for 5 minutes

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.34 - Melt chocolate on stove
0.33 - Take off heat and let cool completely (~20 min)
0.22 - Pour into mug
0.11 - Simmer for 5 minutes
…

P = 80%

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.3 -	* 0.1	ocolate on stove
0.29 -	* 1.0	off heat and let cool completely (~20 min)
0.2 -	* 1.0	to mug
0.1 -	* 1.0	for 5 minutes
…	…	

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.03 - Melt chocolate on stove
0.29 - Take off heat and let cool completely (~20 min)
0.2 - Pour into mug
0.1 - Simmer for 5 minutes
…

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Fifth variant: **Contrastive** sampling

Method: At each step, adjust scores based on similarity with recent outputs, then choose the highest scoring option

Ingredients:

4oz Chocolate, 70% cocoa
1cup Milk

This is starting to mix modelling with inference

Steps:

1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering

0.04 - Melt chocolate on stove
0.40 - Take off heat and let cool completely (~20 min)
0.27 - Pour into mug
0.14 - Simmer for 5 minutes
…

This can be combined with previous approaches

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Greedy variant comparison

Top-1	Argmax
Random	Sample - full distribution
Top-K	Sample - partial list, fixed length
Top-P	Sample - partial list, variable length
Contrastive	Adjust scores, then argmax

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Beam Search

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Core beam idea: Keep track of multiple options

Input at each step:

Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas
-	Noun	Noun Verb	Noun Verb Verb

Output so far [Greedy]:

Noun	Noun Verb	Noun Verb Prep	Noun Verb Prep Noun

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Core beam idea: Keep track of multiple options

Input at each step:

Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas	Fruit flies like bananas
-	Noun	Noun Verb	Adj Noun Verb
	Verb	Adj Noun	Noun Verb Prep
	Adj	Noun Noun	Noun Verb Verb

Output so far [Beam]:

Noun	Noun Verb	Adj Noun Verb	Adj Noun Verb Noun
Verb	Adj Noun	Noun Verb Prep	Noun Verb Prep Noun
Adj	Noun Noun	Noun Verb Verb	Adj Noun Verb Adj

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Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)

A
B
C
D
E

New K best options after doing this step

A, x
A, p
A, g
A, u

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Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)

A
B
C
D
E

New K best options after doing this step

A, x
A, q
B, x
A, g
B, p

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Implement with a list of options at each step that you consider extending

K best options so far (K = 5 here)

A
B
C
D
E

New K best options after doing this step

E, q
A, x
A, q
B, x

Depending on your scoring method, sometimes you can stop early - if you know that none of the remaining options will be better

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For variable length outputs, there are multiple possible beam definitions

Beams based on number of output lines:

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings
Steps:
1. Melt chocolate on stove

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Steps:
1. Melt chocolate on stove
2. Slowly add milk

Beams based on step number:

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings
Steps:
1. Melt chocolate on stove

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Steps:
1. Melt chocolate on stove

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Implementation note - how to select the top K?

A
B
C
D
E

Simple approach, O(nk)
For each option, go through the list to find where it goes. Once found, insert and update.

Heap approach, O(n log k)
Use a min-heap. Update it with each new item.

Quickselect approach, O(n)
Record all options. Use quickselect to find the Kth best. Make one pass through the list to get the other K-1.

Usually k is small enough that any of these are fine and the computation of different options dominates anyway

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Sometimes beam search does not capture useful variation

Ingredients:
4oz Chocolate, 70% cocoa

Ingredients:
4oz Chocolate, 75% cocoa

Ingredients:
3.5oz Chocolate, 75% cocoa

Ingredients:
3.5oz Chocolate, 75% cocoa

Ingredients:
4oz Cocoa Powder

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Graph Search

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We can also treat the problem as a search task and use graph theory

-

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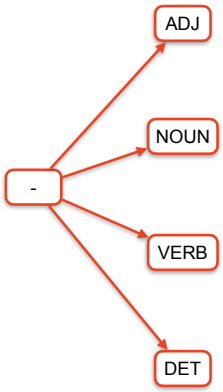

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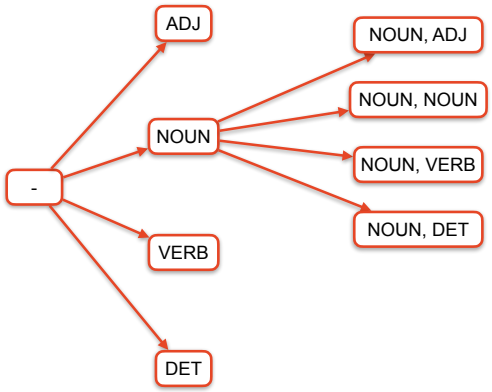

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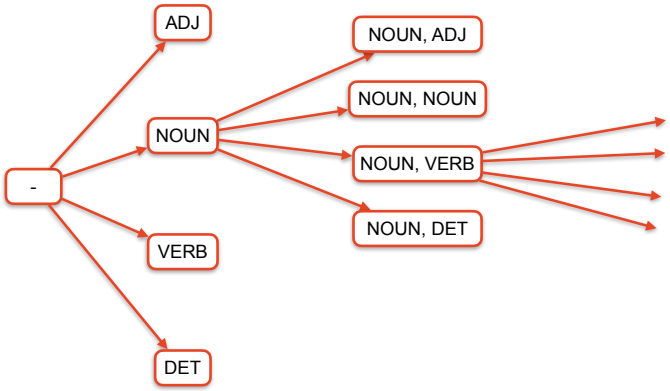

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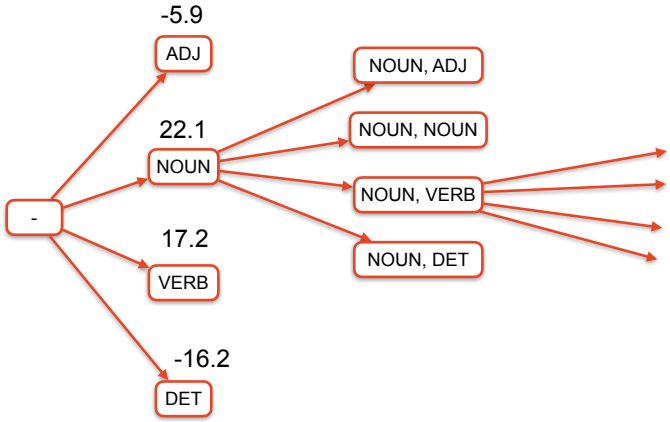
Core idea: Estimate final score and use that to guide generation



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Exhaustive
Greedy
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Graph Search
Dynamic Programming
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Core idea: Estimate final score and use that to guide generation

Ingredients:
4oz Chocolate, 70% cocoa
1cup Milk
Chocolate shavings

Steps:
1. Melt chocolate on stove
2. Slowly add milk
3. Heat until simmering
4. Take off heat and let cool

Score of generation so far

Estimate of score to finish

Note - if you learned about A* search, this sounds similar, but there we are finding the shortest path, here we want the largest score.

3 minute Break - stretch and visit Menti

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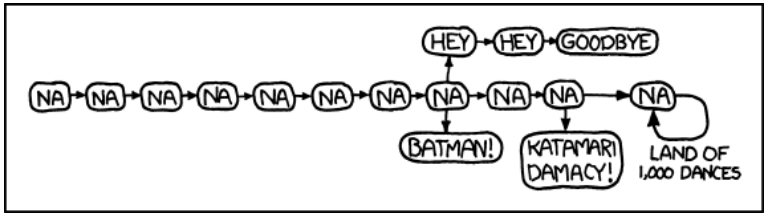
Na



[I hear that there are actual lyrics later on in Land of 1,000 Dances, but other than the occasional "I said," I've never listened long enough to hear any of them.]

Source: https://xkcd.com/851/

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Implement an RNN in PyTorch

Pre-work: Reading and example RNN code

Workshop: Implementing the RNN from lecture 3

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Muddy Card

Open shortly, closes at 7:05pm

https://saipll.shinyapps.io/student-interface/

If you do not wish to participate in the study, use the Ed form instead

Go to Ed → Lessons → Muddy Cards Lecture 4